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Efficient Estimation of Traveltime Upper Bounds Using Fermat's Principle for Enhanced Seismic Imaging

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Abstract

This paper presents a novel method for estimating the maximum traveltime for seismic rays within a given velocity model. Utilizing Fermat's principle, the method calculates traveltimes along hypothetical straight paths from the source to boundary points and selects the maximum value to establish an optimal traveltime upper bound. Numerical tests on synthetic models validate the method's accuracy and efficiency, significantly reducing computational costs while ensuring precision in seismic imaging processes. The proposed method adapts seamlessly to complex topographies and multi-parameter models, making it a versatile tool for various geological scenarios.

Introduction

Seismic imaging is a fundamental technique used in geophysical exploration to map subsurface structures [1]. It involves the generation of seismic waves using sources such as explosives, air guns, or vibrators. These waves propagate through the subsurface and are reflected, refracted, or diffracted by different geological layers. The reflected waves are recorded by an array of sensors called geophones or hydrophones. The recorded data, known as seismic traces, contain information about the subsurface structures.

The next step in seismic imaging is data processing, which involves filtering, stacking, and migration. Migration is a critical process that converts the recorded seismic data into a coherent image of the subsurface. It corrects for the effects of wave propagation and positions the reflectors at their true locations [2]. Accurate traveltime calculation is essential for migration, as it determines the paths and arrival times of seismic waves.

Ray tracing is a common method used to calculate traveltimes [3]. It involves tracing the paths of seismic rays through the subsurface model and computing the traveltimes along these paths. The subsurface model is typically represented by a velocity model, which defines the speed at which seismic waves travel through different geological layers. Ray tracing algorithms simulate the propagation of seismic waves by solving the eikonal equation, which describes the relationship between traveltime and velocity.

Ray tracing can be computationally intensive, especially when dealing with complex geological structures [4]. The accuracy of ray tracing depends on the resolution of the velocity model and the number of rays traced. High-resolution models and dense ray sampling provide more accurate traveltimes but require substantial computational resources. Determining the maximal traveltimes or the number of time steps required for ray tracing is a critical parameter that must be set before the process begins. Overestimating this parameter can lead to excessive computational costs, while underestimating it can result in inaccurate imaging.

The traveltimes upper bound is the maximum time taken by seismic waves to travel from the source to any point within the subsurface model. It is a crucial parameter for ray tracing and seismic imaging. Accurate estimation of the traveltimes upper bound ensures that the ray tracing algorithm covers all possible paths and arrival times, providing reliable traveltimes calculations. An overestimated upper bound leads to unnecessary computational expenses, while an underestimated upper bound compromises the accuracy of the seismic image.

Existing methods for estimating traveltimes upper bounds often involve tracing a limited number of rays with coarsely sampled emergent angles. These methods can provide reasonable estimates but still require substantial computational resources. Moreover, the coarse sampling of emergent angles may fail to account for the complexity of the subsurface model, leading to potential underestimation of the maximum traveltimes. This paper proposes a novel method that addresses these limitations by employing a hypothetical straight path approach based on Fermat's principle.

Methods

Kirchhoff Depth Migration

Kirchhoff depth migration (KDMIG) is a prominent seismic imaging technique that repositions recorded seismic events to their accurate subsurface locations. This method is particularly advantageous in regions with complex geological structures, as it accommodates variations in seismic velocity and effectively handles steeply dipping reflectors.

The general formulation of KDMIG is represented by the following integral expression [5]:

$$I(\xi) = \int W(\xi, g, s) D[t = t_D(\xi, g, s), g, s] dg ds, \quad (1)$$

where $I(\xi)$, defined in a three-dimensional space $\xi = (z_\xi, x_\xi, y_\xi)$, is equal to the integral of the data values $D(t, r, s)$ evaluated at the time $t_D(\xi, g, s)$ and weighted by an appropriate factor $W(\xi, g, s)$. The amplitude factor should also include a one-dimensional convolution by a derivative filter, with the order of the derivative depending on the dimensionality of the integration.

In KDMIG, the time delays $t_D(\xi, g, s)$ are typically computed numerically using ray tracing or an Eikonal equation solver. Figure 1 illustrates how the downgoing and the upgoing rays are traced through an interval velocity function $v(z, x, y)$. The resulting time delays can be expressed as:

$$t_D(\xi, g, s) = t_s[\xi, s, v(z, x, y)] + t_g[\xi, g, v(z, x, y)], \quad (2)$$

where t_s is the time delay from the source location s to the image point ξ , and t_g is the time delay from the same image point ξ to the receiver location g . For a fixed source or a fixed receiver with varied image points, t_s and t_g can be regarded as two traveltimes tables or traveltimes volumes.

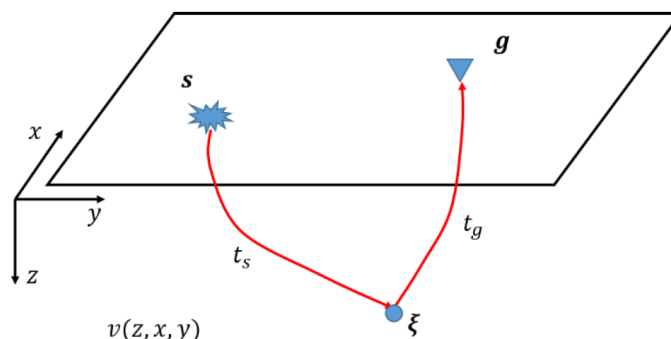


Figure 1—Graphical illustration of the ray tracing necessary to compute the time delays used for KDMIG of a single trace.

There are various implementations of Kirchhoff depth migration, but they generally share the following major steps [6]:

1. Computing the traveltimes from the source and from the receiver to the image points
2. Preprocessing the input seismic traces
3. Migrating the seismic traces according to the traveltimes information
4. Postprocessing and outputting the migration results

Within the KDMIG workflow, the computational cost mainly comes from traveltimes calculation by ray tracing. The ray tracing cost depends on the step number or maximal time of ray tracing. Large estimation of maximal ray tracing time leads to high computational cost, whereas less estimation may cause imaging inaccuracy. To address these issues, the proposed Fermat-principle-based method provides an accurate and efficient estimation of maximal ray tracing time, which can balance the accuracy and efficiency in a solid way.

Traveltimes Upper Bound Estimation with Fermat's Principle

The method presented in this study introduces an efficient and precise approach for estimating the upper bound of traveltimes based on Fermat's Principle. Fermat's Principle posits that a seismic ray traverses between two points along a path that requires stationary time, which is typically the least time, compared to other proximate paths. In a homogeneous medium, this principle implies that the ray follows a straight line, representing the shortest path, as illustrated in Figure 2a. However, in an inhomogeneous medium, the ray's trajectory deviates from a straight line due to variations in the medium's properties. This deviation is depicted in Figure 2b, where the actual ray path, shown as a black solid line, requires less traveltime than the hypothetical straight path, represented by the yellow dashed line. Consequently, according to Fermat's Principle, the traveltime along the true path will not exceed that of the hypothetical straight path, irrespective of the medium's complexity.

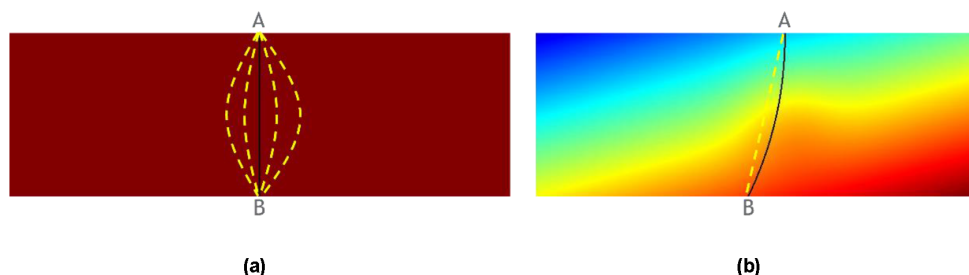


Figure 2—In accordance with Fermat's principle, the ray travels between A and B along the black solid paths requires the least time, as compared to other nearby yellow dash paths. This is demonstrated in (a) homogeneous velocity model and (b) inhomogeneous velocity model.

To elaborate, Fermat's Principle is grounded in the concept of stationary action, which in the context of seismic wave propagation, translates to the path of least time. This principle is fundamental in the field of geometrical optics and has been adapted to seismic ray theory. In a homogeneous medium, where the velocity is constant, the shortest path between two points is a straight line. The traveltime along this path can be easily calculated by dividing the distance by the velocity.

In contrast, in an inhomogeneous medium, where the velocity varies spatially, the ray path bends according to Snell's Law, which governs the refraction of waves. The actual path taken by the ray minimizes the traveltime, considering the varying velocities along the path. This results in a curved trajectory that is more complex to compute. However, Fermat's Principle assures that the traveltime along this curved path will not exceed the traveltime along a hypothetical straight path, which serves as an upper bound.

Procedure for Estimating Traveltime Upper Bound

The proposed method leverages Fermat's Principle by first converting the velocity model into a slowness model, where slowness is the reciprocal of velocity. This transformation simplifies the integration of traveltimes along straight paths from the source to various boundary points within the model. By calculating the traveltimes along these hypothetical straight paths and selecting the maximum value, the method establishes an optimal upper bound for traveltime within the given model. Below delineates the comprehensive procedural sequence for estimating traveltime upper bound:

1. Convert the velocity model into a slowness model: The velocity model represents the speed at which seismic waves travel through the subsurface. By converting this model into a slowness model, which represents the reciprocal of velocity, we can facilitate the integration of traveltimes along straight paths.
2. Integrate slowness along straight paths from the source to boundary points of the model: The integration of slowness along straight paths involves calculating the traveltime for seismic waves traveling directly from the source to various boundary points within the model. This step leverages Fermat's principle, ensuring that the calculated traveltimes along these hypothetical straight paths provide a reliable upper bound.
3. Determine the upper bound for seismic rays traveling within the model by selecting the maximal traveltime value from these paths: By selecting the maximum traveltime value from the calculated traveltimes along straight paths, we establish an optimal upper bound for subsequent ray tracing steps. This upper bound ensures that the traveltime along the actual ray path will not exceed the calculated maximum value.

Figure 3 illustrates the procedure of traveltime upper bound estimation. The independence of slowness integration along various paths allows for parallel algorithm implementation, further enhancing the efficiency of the proposed method.

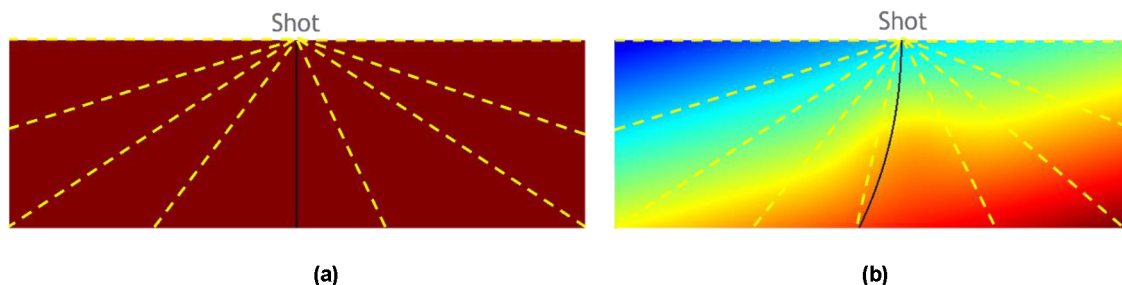


Figure 3—Estimating the traveltime upper bound is illustrated by comparing the travel times from the shot to the boundary points along hypohetic straight paths, applicable in both (a) homogeneous and (b) inhomogeneous models.

Adaptation to Complex Topography and Multi-Parameter Models

In regions with complex topography, the surface boundaries are not flat but exhibit significant variations in elevation. These variations can affect the propagation of seismic waves, leading to deviations in the ray paths. To address this, the method modifies the hypothetical paths from the source to the surface boundaries, ensuring that the traveltimes calculations remain accurate despite the topographical complexities. This modification involves integrating the slowness along non-straight paths that follow the contours of the terrain. By doing so, the method ensures that the estimated traveltime upper bound accurately reflects the true traveltime required for seismic waves to traverse the complex topography.

Additionally, the method extends its applicability to multi-parameter models, such as elastic and anisotropic models, by adjusting the traveltime calculation for each segment. Multi-parameter models are essential in seismic imaging as they provide a more comprehensive representation of the subsurface properties. Elastic models account for the effects of both compressional and shear waves, while anisotropic models consider the directional dependence of seismic wave velocities. These models are crucial for accurately imaging complex geological structures, such as fractured reservoirs and anisotropic formations.

The extension to multi-parameter models involves segmenting the ray paths and calculating the traveltime for each segment based on the specific parameters of the model. For elastic models, the method calculates the traveltime for both compressional and shear waves, ensuring that the upper bound estimation encompasses the effects of both wave types. For anisotropic models, the method adjusts the traveltime calculation to account for the directional variations in wave velocities. This adjustment ensures that the estimated upper bound accurately reflects the true traveltime required for seismic waves to traverse the anisotropic medium.

By adapting to complex topographies and extending to multi-parameter models, the proposed method enhances its versatility and applicability in various geological scenarios. It provides a robust framework for estimating traveltime upper bounds, ensuring accurate and efficient seismic imaging across different environments. The ability to handle complex topographies and multi-parameter models makes the method a valuable tool for geophysical exploration, offering reliable traveltime estimations that improve the quality and accuracy of seismic images.

Results

The efficacy of the proposed methodology was validated through two complex synthetic models. Comparative analysis between true and estimated traveltime upper bounds at various source locations has been conducted to demonstrate the method's efficiency and accuracy.

SEG/EAGE Salt Model

The SEG/EAGE salt model is a widely used synthetic model in geophysical research, known for its complex geological structures [7]. The model includes intricate salt body geometries that pose significant challenges for traveltime calculation. To validate the proposed method, we conducted numerical tests using this model.

As shown in [Figure 4](#), The shot location was positioned at the intersection of blue lines on the model's surface. The true maximal traveltime was calculated by performing ray tracing and identifying the maximum traveltime from the traveltime field. This value was marked by a blue cross in the traveltime field.

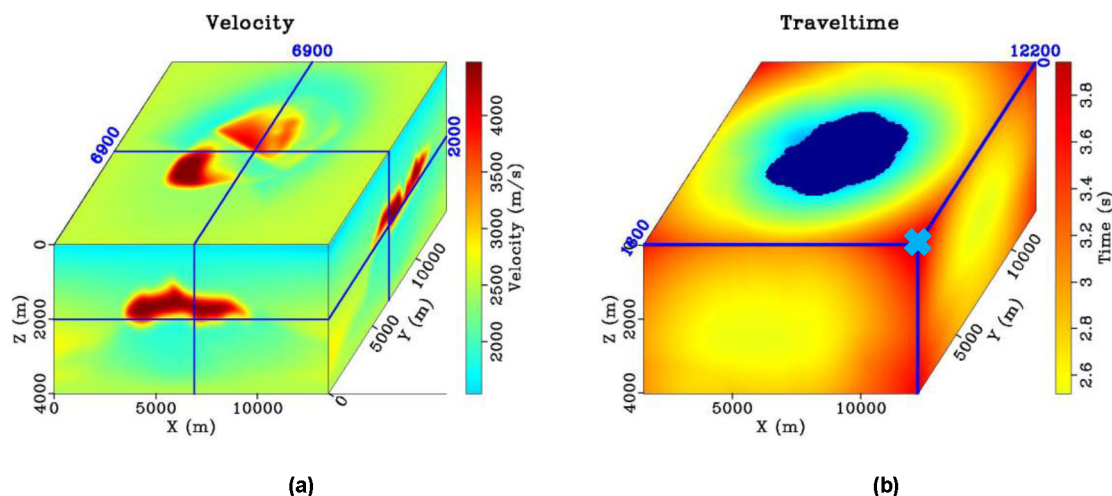


Figure 4—Tests on SEG/EAGE salt model. (a) Velocity model, the shot is located at the intersection of blue lines on the surface; (b) Calculated traveltimes field from the shot to the subsurface grid points, with the maximal traveltimes indicated by a blue cross.

While ray tracing can yield accurate traveltimes upper bounds, it requires significant computational resources. To evaluate the efficiency of the proposed method, we compared the true upper bounds obtained through ray tracing with the estimated upper bounds calculated using the hypothetical straight path approach. Figure 5 presents the comparative analysis, demonstrating that the proposed method provides reasonable estimates with minimal computational overhead.

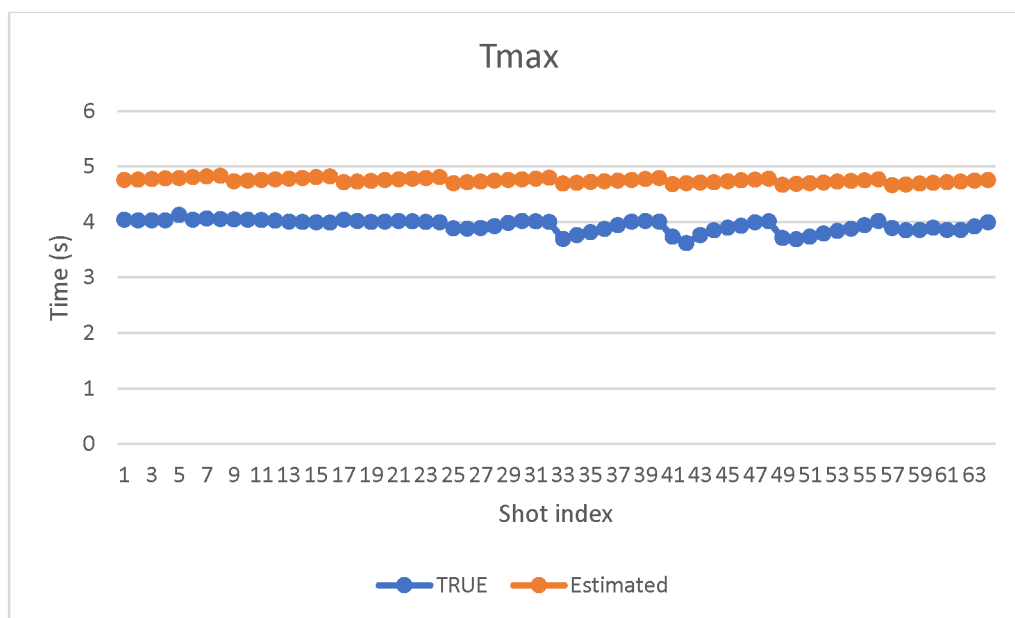


Figure 5—Comparisons of the true and estimated upper bounds of travel times. The true upper bounds are obtained through ray tracing, while the estimated upper bounds are calculated by the proposed method. The shots are located at various locations on the surface of the model.

SEAM Arid Model

The SEAM Arid model is another synthetic model commonly used in geophysical research, designed to simulate desert environments near the surface [8]. This model includes complex land structures that challenge traveltimes calculation.

As shown in Figure 6, the shot location was positioned at the intersection of blue lines on the model's surface. The true maximal traveltime was calculated by performing ray tracing and identifying the maximum traveltime from the traveltime field. This value was marked by a blue cross in the traveltime field.

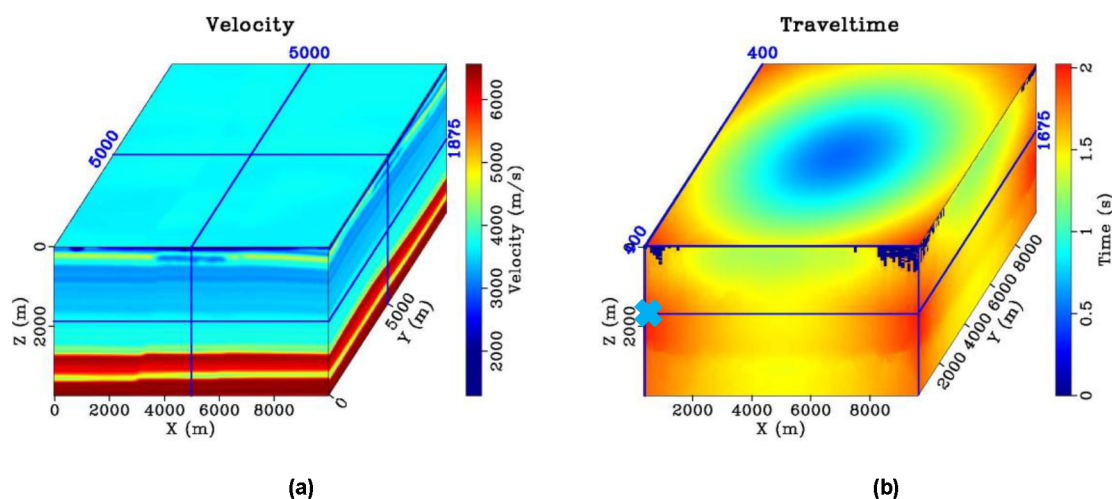


Figure 6—Tests on SEAM Arid model. (a) Velocity model, the shot is positioned at the intersection of blue lines on the surface; (b) Calculated traveltime field from the shot to the subsurface grid points, with the maximal traveltime indicated by a blue cross.

Like the SEG/EAGE salt model, ray tracing can accurately determine upper bounds for traveltimes but is computationally intensive. To evaluate the efficiency of the proposed method, we compared the true upper bounds obtained through ray tracing with the estimated upper bounds calculated using the hypothetical straight path approach. Figure 7 presents the comparative analysis, demonstrating that the proposed method provides reasonable upper bound estimations with negligible computational cost.

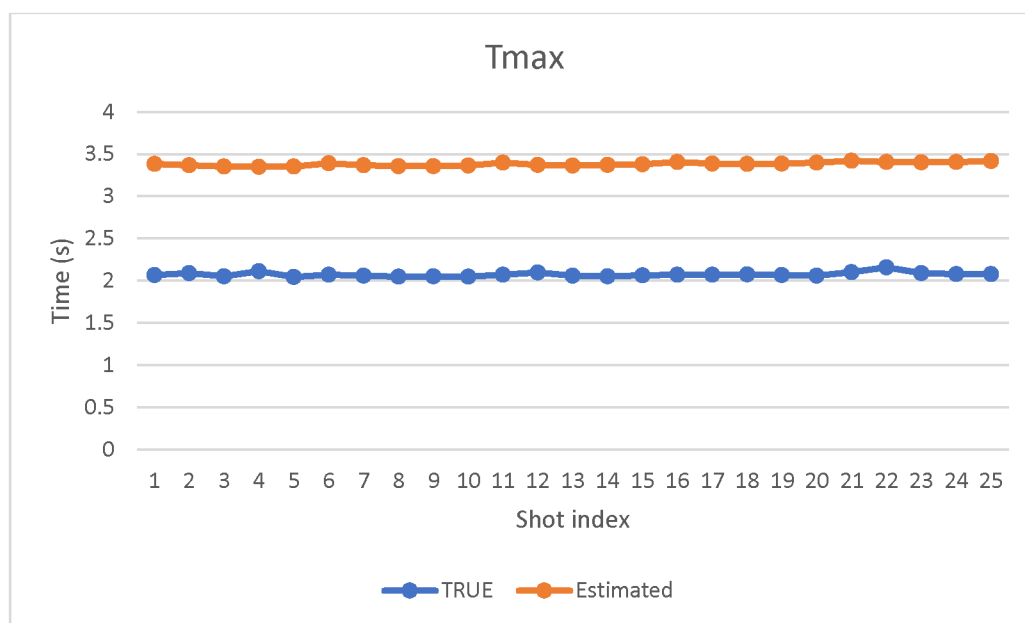


Figure 7—Comparisons of the true and estimated upper bounds of travel times. The true values are calculated via ray tracing, while the estimated values are calculated by the proposed method. The shots are located at various locations on the surface of the model.

Discussions

Performance and Efficiency

The proposed method significantly reduces computational costs associated with traveltime calculation in seismic imaging. By employing hypothetical straight paths, the method eliminates the need to trace actual ray paths, ensuring efficiency without compromising accuracy. The method's adaptability to complex topographies and multi-parameter models further enhances its applicability in various geological scenarios.

Limitations

The method may overestimate the upper bound of traveltime when the velocity model exhibits significant variations, potentially leading to unnecessary computational expenses. Future work should focus on refining the method to address this limitation.

Comparison with Existing Methods

Existing methods for estimating traveltime upper bounds often involve tracing a limited number of rays with coarsely sampled emergent angles. While these methods can provide reasonable estimates, they still require substantial computational resources. Moreover, the coarse sampling of emergent angles may fail to account for the complexity of the subsurface model, leading to potential underestimation of the maximum traveltime.

In contrast, the proposed method eliminates the need to trace actual ray paths by employing a hypothetical straight path approach based on Fermat's principle. This approach significantly reduces computational demands while ensuring that the calculated upper bound of traveltime surpasses the traveltime at any other grid point within the model. The method systematically evaluates all boundary points, ensuring comprehensive coverage and accurate upper bound estimation.

Conclusions

This paper presents a fast and accurate method for estimating the traveltime upper bound based on Fermat's principle. The proposed method leverages the concept that the traveltime along the actual path must not exceed that of a hypothetical straight path, ensuring that the upper bound estimation is both reliable and efficient. Numerical tests conducted on synthetic models, such as the SEG/EAGE salt model and the SEAM Arid model, validate the method's capability to deliver precise upper bound estimations with minimal computational costs. These tests demonstrate that the method provides reasonable estimates of the traveltime upper bound, significantly reducing the computational overhead compared to traditional ray tracing methods. Its validated accuracy, adaptability to complex topographies, and applicability to multi-parameter models make it a valuable tool for geophysical exploration. Future research should focus on further refining the method to address any limitations and enhance its applicability to an even broader range of geological conditions. The development of this method represents a significant advancement in the field of seismic imaging, promising improved efficiency and precision in geophysical exploration.

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